Enterprise AI Test, Evaluation & Assurance Strategy

Unifying LLM, Traditional ML, AI-Harness, and AI-Enabled-System Evaluation Under One Capability

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Executive Summary

Today, AI evaluation is fragmented: one group benchmarks LLMs, another validates translation or NER models, another red-teams agentic coding tools, another runs operational tests on fielded systems. Each maintains its own datasets, metrics, and reports. The result — duplicated effort, incomparable scores, and no enterprise-level answer to "is this AI system ready to deploy" — mirrors exactly why 70–90% of enterprise AI initiatives fail to reach production value (Gartner, 2025) and why only 3% of organizations report >10% ROI from AI (McKinsey *State of AI*, 2025). This is not a novel problem: NIST has run rigorous, shared benchmark programs for speech recognition (OpenASR/OpenSAT) and machine translation (Open MT, WMT) for over two decades, proving that centralized, community-grade evaluation outperforms fragmented, per-team testing at enterprise scale. **This strategy extends that same discipline to LLMs, traditional ML, AI harnesses, and AI-enabled systems under one governed capability**, built on established public frameworks rather than invented from scratch.

Grounding in Established Practice

Rather than build a bespoke methodology, this strategy adopts and integrates the leading public frameworks already governing each layer of the stack:

Layer	Adopted framework / precedent	What it contributes
Governance	**NIST AI RMF** (GOVERN–MAP–MEASURE–MANAGE) + GenAI Profile (NIST-AI-600-1)	Common risk taxonomy and lifecycle vocabulary
Management system	**ISO/IEC 42001** (AIMS, PDCA)	Auditable, certifiable management-system structure
LLM/agent evaluation	**UK AISI Inspect**, **Stanford HELM**, **Chatbot Arena** (Bradley-Terry, Chiang et al. 2024)	Open eval harnesses, multi-metric scoring, pairwise ranking
Agentic-harness evaluation	**METR** time-horizon methodology, **SWE-bench**, **Terminal-Bench**, HAL (Princeton, 2025)	Multi-trial, cost/accuracy-aware harness comparison — same scaffolding vs. model distinction this org's own AICO platform already implements
Traditional ML	**NIST OpenASR/OpenSAT** (WER), **WMT/Open MT** (BLEU/COMET), CoNLL-style F1 (NER)	Decades-proven, modality-specific metric standards — not reinvented
Security	**MITRE ATLAS**	Adversarial-ML threat taxonomy for red-teaming and risk exposure

Layer	Adopted framework / precedent	What it contributes
Acquisition/operational T&E	**DoDI 5000.89** + CDAO AI T&E Framework	Model → integration → operational → live-fire lifecycle for fielded AI-enabled systems
Production assurance	LLMOps/MLOps continuous-evaluation and drift-monitoring practice	Post-deployment, not just pre-deployment, evaluation
Federal precedent	**EO 14409** (AI Innovation & Security) + **EO 14413** (Quantum Innovation), both Jun. 2026	Both direct a dedicated national assessment body (NSA/CISA/NIST classified AI-model benchmarking; a national quantum-performance-assessment center) rather than leaving evaluation to individual developers — precedent, at the highest level of government, for evaluation as its own capability

Strategic Objectives

1. **Standardize** — one metric catalog, one benchmark/dataset registry, one report format, mapped to a shared *Enterprise Capability Taxonomy* (reasoning, coding, translation, NER, speech, vision, forecasting, harness quality, system/mission outcomes) so scores stay comparable as underlying models churn.
2. **Evaluate systems, not just models** — a translation → NER → retrieval → LLM → decision pipeline is what the business actually deploys; score the pipeline and the mission outcome, not only each component in isolation.
3. **Separate harness from model** — per the Tessl 2026 position paper, the same model can swing 20+ points across four agent harnesses on identical tasks; harness quality (Claude Code, Codex CLI, and similar tools) must be tracked as its own evaluated capability, not conflated with model capability.
4. **Reuse before build** — shared infrastructure, benchmark/dataset registries, and an automated orchestration platform (submit → auto-select benchmark suite → execute → score → report) rather than per-team pipelines.
5. **Continuous assurance** — extend evaluation past a pre-deployment gate into production drift monitoring and scheduled re-certification, consistent with DoDI 5000.89's full lifecycle and modern LLMOps practice.

Operating Model

Governance sets standards and risk tolerance; it does not run evaluations — preserving the independence NIST AI RMF and DoDI 5000.89 both require between builders and evaluators.

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                     AI Assurance Governance Board
                  (standards, risk tiers, certification policy)
                                 |
                   Evaluation Orchestration Platform
        (submit -> select benchmarks -> execute -> score -> report)
                                 |
   ---------------------------------------------------------------
   |            |              |              |                 |
Benchmark    Dataset      Model/Harness    AI-Enabled       Operational &
Engineering  Engineering  Evaluation      System Eval     Continuous Monitoring
   ---------------------------------------------------------------
                                 |
                   Executive Scorecards & Deployment Evidence
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Four evaluation capabilities, one taxonomy: **Model** (LLM, translation, NER, speech, vision — modality-appropriate metrics), **Harness** (agentic tool quality, isolated from the model it drives — Bradley-Terry/Elo leaderboards plus GLMM variance decomposition to attribute outcomes to harness vs. model vs. task), **AI-Enabled System** (the full pipeline/agent workflow a user actually touches), and **Operational** (stress, adversarial input, degraded data, mission effectiveness — DOT&E-style OT&E).

Risk-Tiered Depth

Not every model needs full operational testing. Depth scales with deployment risk (NIST AI RMF's MAP function, DoDI 5000.89's tailoring principle):

Tier	Scope	Adds on top of the tier below
1	Mission-critical / customer-facing	Operational test, red team (MITRE ATLAS), continuous monitoring, re-certification
2	Enterprise-facing system	AI-enabled system & harness evaluation
3	Departmental tool	Independent dataset validation
4	Research / experimental	Benchmark suite (baseline)

Roadmap

Year 1 — Foundation & Integration: stand up the governance board, benchmark/dataset registries, and common metrics catalog mapped to the capability taxonomy; consolidate existing LLM, translation/NER/speech, and harness evaluation efforts under shared standards without disbanding domain expertise; bring the orchestration platform live; standardize AI-enabled system and harness evaluation (Bradley-Terry leaderboards, cost/Pareto frontier) enterprise-wide; replace per-team reports with executive scorecards. **Year 2 — Operational & Continuous:** operational test scenarios, red-teaming, and production drift monitoring/re-certification become the default lifecycle for Tier 1–2 systems, closing the loop from field evidence back into the benchmark and dataset registries.

A Working Instantiation

The organization's own **AICO (AI Capability Observatory)** platform already implements the core of this architecture — a hierarchical Bayesian scoring engine for models, a Bradley-Terry/Glicko-2/GLMM harness-comparison layer that keeps harness quality separate from model quality, and MITRE ATLAS-mapped risk exposure — and is the natural first system of record for the Model and Harness evaluation capabilities above.

Sources

NIST AI RMF · ISO/IEC 42001 · DoDI 5000.89 · CDAO AI T&E Framework · UK AISI Inspect · Stanford HELM · MITRE ATLAS · NIST OpenASR · NIST Open MT · EO 14409 · EO 14413

Chiang et al., *Chatbot Arena* (arXiv:2403.04132) · METR *Time-Horizon* methodology · Tessl, *Position: Coding Benchmarks Are Misaligned with Agentic Software Engineering* (arXiv:2606.17799) · *Holistic Agent Leaderboard* (arXiv:2510.11977) · Gartner (2025 AI project failure analysis) · McKinsey, *The State of AI* (2025)